

ARGHADIP DAS

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Education

Purdue University

Jan 2022 – Dec 2026 (Expected)

M.S. & Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0

West Lafayette, United States

Courses: GPU Architecture, Deep Learning, Advanced VLSI Design, Computer Architecture, Operating Systems, SoC Design

Jadavpur University

Aug 2016 – Aug 2020

B.E. in Electronics and Telecommunication Engineering | GPA: 9.87/10 (Gold Medalist)

Kolkata, India

Experience

Intel | AI Solutions Engineering PhD Intern

Santa Clara, USA | May 2026 – Aug 2026

- *Proposed the QPU (Quality-Configurable NPU)*, instrumenting the entire accelerator datapath with runtime-tunable approximation knobs to accommodate diverse model families (CNNs, transformers, MoE, and SSMs).
- *Developed a joint tuning framework* that exploits workload error resilience on the fly, recovering unclaimed energy savings to achieve *over 10% energy savings within 1% of baseline quality* from INT4 to FP16.

Intel | Client AI Research Intern

Santa Clara, USA | May 2024 – Dec 2024

- Mapped and optimized emerging *graph, transformer-free and hybrid neural networks on Intel Neural Processing Unit (NPU)*, achieving *over 2X to 17X performance improvement* compared to the CPU.
- *Developed patentable innovations in efficient Graph Neural Networks (GNNs) and transformer-free network mapping*, submitting *4 patents*, 3 conference articles and 1 journal article for publication.

Texas Instruments | Validation/Test Engineer

Bangalore, India | Aug 2020 – Dec 2021

- *Led a 4-person team in validating and characterizing the DP83812 automotive 100BASE-T1 Ethernet PHY*, ensured the timely sign-off of the product through rigorous testing and issue resolution using *LabVIEW and TestStand*.
- Achieved a *12X reduction in testing and device characterization time* by designing PCBs using *Altium Designer* for a multi-site setup of the DP83867A industrial PHY (10BASE-Te, 100BASE-TX, and 1000BASE-T).

Texas Instruments | Design Verification (DV) Intern

Bangalore, India | May 2019 – Jul 2019

- *Reduced DV environment development time by 2X* for future products by unifying Ethernet Bus Functional Models (BFMs) and establishing a regression testing baseline.
- *Learned a new language, Specman e Hardware Verification Language (HVL), in 1 week* and demonstrated profound expertise in developing the DV environment using it.

IIT Kharagpur | RTL Design Intern

Kharagpur, India | May 2018 – Jun 2018

- Created a Snort-based Network Intrusion Detection System (NIDS) to *detect and mitigate security attacks with 80% less resources (LUTs) on FPGA*.

Publications

- [1] "NEXUS: Speeding Up the Next Wave of AI Workloads—GNN and SSM Optimization on NPUs," in TECS, 2026.
- [2] "DYNNER: Optimizing Dynamic Power Consumption in Deep Learning Accelerators," in TCASAI, 2026.
- [3] "COSMOS: Designing Energy-Efficient Context-Aware Multimodal Cognitive Systems," in ISLPED, 2026.
- [4] "GraNNite: Enabling High-Performance Execution of GNNs on Resource-Constrained NPUs," in IJCNN, 2025.
- [5] "XAMBA: Enabling Efficient State Space Models on Resource-Constrained NPUs," in SCOPE ICLRW, 2025.
- [6] "Towards Efficient Acceleration of Hyena and Kolmogorov–Arnold Networks on NPUs," in ICEdge, 2025.
- [7] "LLM-NPU: Towards Efficient Foundation Model Inference on Low-Power Neural Processing Units," in COINS, 2025.
- [8] "SparseDroop: Hardware-Software Co-design for Mitigating Voltage Droop in DNN Accelerators," in JLPEA, 2025.
- [9] "Towards Energy-Accuracy Scalable Multimodal Cognitive Systems," in IEEE ESL, 2024.
- [10] "Toward Energy-Efficient Collaborative Inference Using Multisystem Approximations," in IEEE IoTJ, 2024.

Technical Skills

Languages: CUDA, Python, SystemVerilog, Verilog, C++, C, Specman e HVL, MIPS, MicroPython, L^AT_EX

Developer Tools: Quartus, Questasim, Vivado, Xilinx ISE, Virtuoso, Altium Designer, MATLAB, LabVIEW, TestStand

Hardware Platforms: FPGAs (Stratix IV, Cyclone IV), NVIDIA Jetson Nano, ESP32 μ C, Raspberry Pi, Arduino

Accomplishments

ECE STAR Fellowship: Granted 5-year Purdue ECE Star Fellowship as a top Ph.D. applicant.

GATE Topper: Secured All India Rank 5 in GATE (Graduate Aptitude Test in Engineering) for EC in 2020.